MEDICAL INSURANCE COST PREDICTION USING MACHINE LEARNING

A PROJECT REPORT

Submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in Computer Science and Engineering

Data Science

Under the guidance of

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Submitted by

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BRAINWARE UNIVERSITY

(A State Private University established under the Brainware University Act, 2015) In association with **Ardent Computech Pvt. Ltd**



Submission Year: 2026

CERTIFICATE

This is to certify that the project report entitled **"Medical Insurance Cost Prediction Using Machine Learning"** is a record of bona fide work carried out by **SOYEL MONDAL** (Reg. No.: **INT/ASS/BU/23633**) under my guidance at the Ardent Computech Pvt. Ltd during the period from August 2026 to September 2026. The report is submitted in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering Data Science** of Brainware University.

To the best of my knowledge, the work embodied in this report is original, has been carried out by the student herself, and has not been submitted to any other university or institution for the award of any degree or diploma. The report is worthy of acceptance in its present form.



Mr. Tridip Kundu
Project Guide Data Science Faculty
Ardent Computech Pvt. Ltd
Date: 13/10/2026

DECLARATION

I, **SOYEL MONDAL** (Reg. No.:INT/ASS/BU/23633), hereby declare that the project work presented in this report, entitled **"Medical Insurance Cost Prediction Using Machine Learning"**, submitted in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering Data Science**, is an authentic record of the work carried out by me under the guidance of Mr. Tridip Kundu, Data Science Faculty, Ardent Computech Pvt. Ltd.

I further declare that this work is original, that it has not been copied from any other source without proper acknowledgement, and that the matter embodied in this report has not been submitted elsewhere for the award of any degree, diploma or certificate to the best of my knowledge and belief.

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I am thankful to the faculty members and academic staff of Brainware University for providing the learning environment, resources and support required to complete this project. I also acknowledge the Ardent Computech Pvt. Ltd for the training and practical exposure that supported the implementation of the predictive system.

I would also like to thank the open-source community and the maintainers of the dataset and software libraries used in this project. The availability of the Medical Cost Personal Dataset and tools such as Python, Pandas, NumPy, Matplotlib and Scikit-learn made it possible to conduct a reproducible study on medical insurance cost prediction.

Finally, I express my gratitude to my family, friends and colleagues for their encouragement and patience during the completion of this work.

ABSTRACT

This project presents an end-to-end machine learning system for predicting annual medical insurance charges from demographic and lifestyle attributes. The study uses the Medical Cost Personal Dataset containing 1,338 records and seven columns: age, sex, bmi, children, smoker, region and charges. A data-quality audit found no missing values and one exact duplicate pair; the source dataset was retained so that the modelling results remain directly tied to the supplied observations. Exploratory analysis showed a strongly right-skewed target distribution and a pronounced separation in charges between smokers and non-smokers, with an especially strong divergence among individuals who are both smokers and have BMI values of at least 30 kg/m². Based on this pattern, the project engineers a binary interaction feature, obese_smoker, within the prediction pipeline. The final model is an Ordinary Least Squares Linear Regression estimator. Categorical variables are one-hot encoded while numeric variables are passed through in their original scale. An 80:20 train-test split with random_state=42 is used for evaluation. The baseline six-feature OLS model obtains an R^2 of 0.7836 and an MAE of USD 4,181.19. Adding obese_smoker improves the holdout performance to $R^2 = 0.8720$, MAE = USD 2,486.32 and RMSE = USD 4,458.27. The trained pipeline is serialized with joblib and integrated with a Streamlit interface that accepts raw user inputs and returns an estimated annual charge together with contextual risk information. The project demonstrates how targeted feature engineering can substantially improve an interpretable linear model without introducing a complex black-box estimator.

Keywords: Medical insurance, cost prediction, OLS linear regression, feature engineering, obese_smoker, Streamlit, scikit-learn.

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CHAPTER 1

INTRODUCTION

1.1 Background of the Project

Medical insurance systems must estimate future healthcare expenditure from the information available about a policyholder. A cost estimate is useful for analytical planning, risk assessment, scenario analysis and educational demonstrations of actuarial modelling. Because healthcare charges are influenced by several characteristics at the same time, the relationship between an individual profile and the final billed amount is not adequately described by a single rule.

Machine learning provides a systematic way to learn relationships from historical records. In regression problems, the objective is to estimate a continuous target variable from a set of input features. This project studies annual medical charges using six observed profile attributes: age, sex, body mass index (BMI), number of covered children, smoking status and residential region. The work intentionally uses Ordinary Least Squares (OLS) Linear Regression so that the resulting model remains transparent and each learned coefficient can be inspected directly.

1.2 Problem Definition

The problem is to estimate the annual medical charge billed for an individual from the six available input attributes while keeping the modelling process reproducible and interpretable. The model must also account for an important interaction observed during exploratory analysis: high charges are particularly concentrated among records that combine smoking with BMI values at or above 30 kg/m².

Formally, given a feature vector x = {age, sex, bmi, children, smoker, region}, the system produces an estimated continuous target $\hat{y}$ = charges. The final implementation adds the engineered binary variable obese_smoker before categorical encoding and OLS prediction.

1.3 Objectives

1. Inspect the dataset programmatically and verify its shape, schema, missing values and duplicate records.
2. Perform exploratory data analysis to understand the distribution of charges and the relationships between the target and the available predictors.
3. Identify a compact, domain-motivated interaction feature that improves the ability of a linear model to represent the observed high-cost segment.
4. Build and evaluate an OLS Linear Regression model on a fixed 80:20 train-test split using reproducible preprocessing.
5. Compare the baseline six-feature model with the enhanced model containing obese_smoker using MAE, MSE, RMSE and R^2.
6. Persist the fitted prediction pipeline and integrate it with a Streamlit interface for raw-input inference.

1.4 Scope of the Project

The project covers the complete workflow from dataset inspection through model training, evaluation, model serialization and interactive prediction. It is restricted to the supplied 1,338-row dataset and does not attempt to predict diagnoses, future clinical events or exact premiums. The application is designed as an educational and analytical demonstration rather than a production underwriting system.

1.5 Motivation

The main motivation is to show that a transparent linear model can become substantially more useful when the structure of the data is understood before training. Instead of increasing model complexity through a large collection of nonlinear estimators, the project focuses on one clearly motivated interaction. This approach creates a direct link between EDA findings, model design, measured performance and the final user-facing application.

CHAPTER 2

LITERATURE REVIEW

2.1 Existing Systems

Medical cost estimation has traditionally relied on actuarial rating structures, statistical regression and grouped risk factors. More recent analytical systems use machine learning to estimate expenditure, identify high-cost members and support scenario analysis. The central design question is not simply which method can minimize prediction error, but how the method should balance flexibility, stability and interpretability for a domain in which the reasoning behind an estimate can matter as much as the estimate itself.

Linear regression remains a strong baseline for tabular datasets because its coefficients are directly interpretable and its implementation is computationally inexpensive. Categorical predictors can be represented with indicator variables, while interaction terms can be added deliberately when the data show a specific combined effect. Tree ensembles and kernel methods offer greater nonlinear flexibility, but they are not used as the final estimator in this project because the project objective prioritizes transparent OLS modelling.

2.2 Research Context

The literature reviewed for this project covers healthcare cost prediction, regression methodology, feature representation and interpretable statistical learning. Table 2.1 summarizes the sources most relevant to the design choices used in the project.

Table 2.1: Summary of related literature

S. No.	Source	Topic / Method	Relevance to the Project
1	Bertsimas et al. (2008)	Algorithmic prediction of health-care costs	Supports the use of data-driven cost prediction for healthcare expenditure.
2	James et al. (2013)	An Introduction to Statistical Learning	Provides foundations for regression, model evaluation and feature relationships.
3	Hastie, Tibshirani & Friedman (2009)	Statistical learning theory	Provides mathematical context for regression, model flexibility and diagnostics.
4	Hoerl & Kennard (1970)	Ridge regression	Provides context for regularized linear modelling, although Ridge is not the final model here.
5	Friedman (2001)	Gradient boosting	Represents nonlinear ensemble methods considered in the wider modelling landscape but intentionally excluded from the final estimator.
6	Breiman (2001)	Random forests	Provides context for tree ensembles and their nonlinear modelling capability.
7	Drucker et al. (1997)	Support Vector Regression	Provides context for kernel-based regression alternatives.

The project therefore uses the literature as methodological context rather than as a claim that one algorithm is universally superior. The final model choice follows the project requirement for an interpretable linear estimator and is evaluated empirically on the supplied dataset.

2.3 Limitations of Current Approaches

1. Purely additive linear models can miss a combined effect when the influence of one variable depends strongly on another. The smoker-obesity split observed in this dataset is an example.
2. Automated feature generation can produce large, difficult-to-interpret models. A smaller, evidence-driven feature set is easier to document and audit.
3. Complex ensemble models may improve predictive accuracy but generally require additional explanation layers to communicate how an individual estimate was produced.
4. Outlier rules used without domain context can remove valid high-cost observations. For a cost target, extreme values can represent an important part of the distribution rather than corrupt records.
5. Public benchmark datasets are limited compared with real claims systems. They do not contain medical history, diagnoses, longitudinal utilization, provider-level information or policy details.

How this project addresses the gap

The project addresses these limitations by keeping the model linear, engineering one explicitly motivated interaction, retaining the complete target distribution, separating training and testing data with a fixed random seed, and placing the transformation and estimator in a reusable pipeline. The same raw input schema used in training is passed to the saved model at inference time, reducing the risk of preprocessing inconsistencies.

CHAPTER 3

SYSTEM ANALYSIS

3.1 Requirement Analysis

Requirement analysis identifies what the system must accept, process and return. The solution is a supervised regression application with two operational contexts: model development by the analyst/ developer and prediction requests from an end user. During development, the system loads the CSV dataset, performs data checks, explores distributions and trains the model. During inference, the system receives a raw six-feature profile and returns a predicted annual charge.

3.2 Functional Requirements

- The system shall load the supplied insurance.csv dataset and verify its structure.
- The system shall report missing values and duplicate records without silently modifying the raw source data.
- The system shall construct the obese_smoker feature from BMI and smoking status.
- The system shall encode categorical variables consistently with the fitted preprocessing schema.
- The system shall train and evaluate baseline and enhanced OLS models on the same holdout partition.
- The system shall persist the fitted prediction pipeline and use it to generate estimates for new profiles.
- The Streamlit interface shall expose the six required user inputs and display the prediction plus contextual risk information.

3.3 Non-Functional Requirements

- Interpretability: the final estimator should expose explicit coefficients and a clear modelling equation.
- Reproducibility: data splitting and processing should be deterministic under the documented random state and feature rules.
- Performance: training and single-row inference should execute comfortably on a standard laptop.
- Usability: the prediction interface should communicate the result clearly without requiring the user to understand the underlying code.
- Portability: the project should use standard Python libraries and a serialized model artifact that can be loaded by a Python application.

3.4 Feasibility Study

Technical feasibility is high because the dataset is small and the solution depends on mature Python libraries. Economic feasibility is also high because the development stack is based on open-source tools and the dataset is publicly available. Operational feasibility is supported by the saved model pipeline and Streamlit front end, which separates model inference from the notebook-based analysis.

Category	Requirement
Hardware	Laptop or desktop with at least 4 GB RAM; no GPU is required for OLS training.
Operating system	Windows, Linux or macOS.
Language	Python 3.x.
Libraries	Pandas, NumPy, Matplotlib, scikit-learn, joblib and Streamlit.
Environment	Jupyter Notebook / Google Colab for analysis; local Python environment for app execution.
Data source	Kaggle Medical Cost Personal Dataset (insurance.csv).

3.5 Dataset Description

The dataset contains 1,338 individual records and seven columns. Six columns are used as predictors and charges is the continuous target. The data are stored in a single CSV file, making the project suitable for a compact tabular machine learning workflow. The observed ranges and semantic roles of the columns are summarized in Table 3.2.

Attribute	Type	Description
age	Integer	Age of the primary beneficiary in years.
sex	Categorical	Biological sex category: female or male.
bmi	Float	Body mass index in kg/m².
children	Integer	Number of covered dependent children.
smoker	Categorical	Smoking status: yes or no.
region	Categorical	US residential region: northeast, northwest, southeast or southwest.
charges	Float	Individual medical charges billed by the insurer; target variable.

Table 3.2: Attributes of the dataset

A sample of the supplied dataset is consistent with the seven-column schema. The project treats the CSV as the source of truth rather than manually entering benchmark values.

3.6 Data Quality and Statistical Profile

The programmatic audit found zero missing values in every column and one exact duplicate pair. The duplicate was retained in the working dataset so that the modelling workflow remains tied to all 1,338 supplied observations. This is different from deleting suspicious records automatically; the project did not find evidence that the row was corrupt or invalid.

Statistic	age	bmi	children	charges
Count	1,338	1,338	1,338	1,338
Mean	39.2070	30.6633	1.0949	$13,270.42
Std. Dev.	14.0499	6.0981	1.2054	$12,110.01
Minimum	18	15.96	0	$1,121.87
25%	27	26.2962	0	$4,740.29
Median	39	30.4000	1	$9,382.03
75%	51	34.6937	2	$16,639.91
Maximum	64	53.13	5	$63,770.43

Table 3.3: Descriptive statistics of the target and numerical variables

The charge distribution is strongly right-skewed. The mean is approximately USD 3,888 higher than the median, and the upper tail extends to more than USD 63,000. This structure becomes important later because a model that fits only the central mass can perform poorly on high-cost observations.

CHAPTER 4

SYSTEM DESIGN

4.1 Architecture Diagram

The proposed system is organised as a sequential data-to-prediction pipeline. The raw CSV is audited and explored, the obesity-smoking interaction is derived, the data are encoded for regression, and the enhanced OLS estimator is evaluated on an untouched holdout partition. The fitted pipeline is serialized and consumed by the Streamlit application. The design keeps training transformations and inference transformations inside the same logical model artifact.

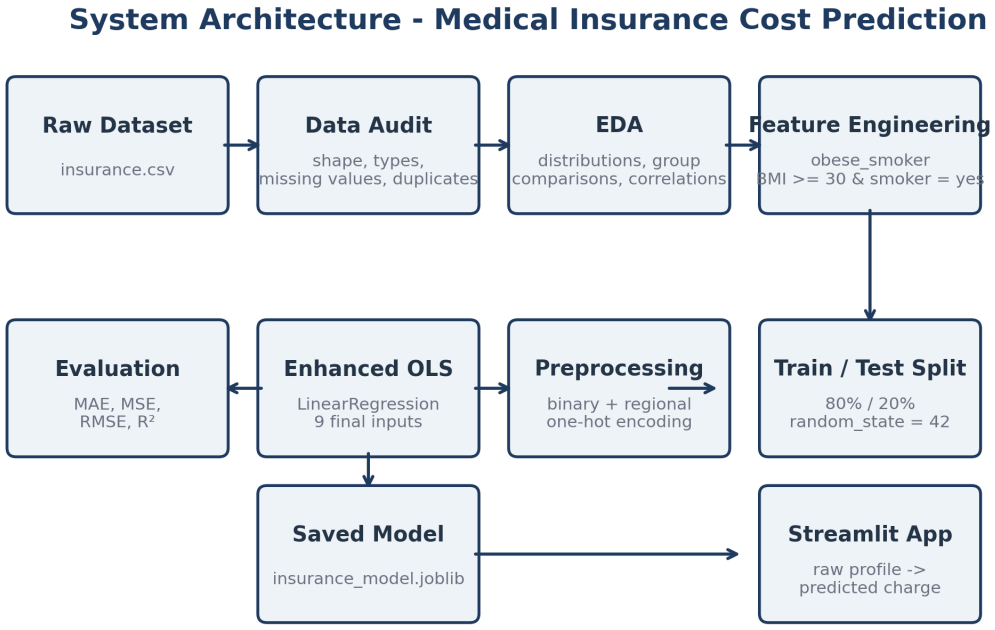


Figure 4.1: System architecture of the medical insurance cost prediction system

4.2 Data Flow Diagram (DFD)

At the context level, the system receives the training dataset from the analyst/developer and a policyholder profile from the end user. The model development path produces metrics and a saved model artifact, while the inference path returns an estimated annual medical charge and contextual risk information. Because the project is a file-based application, the CSV and joblib artifact act as the principal data stores rather than a relational database.

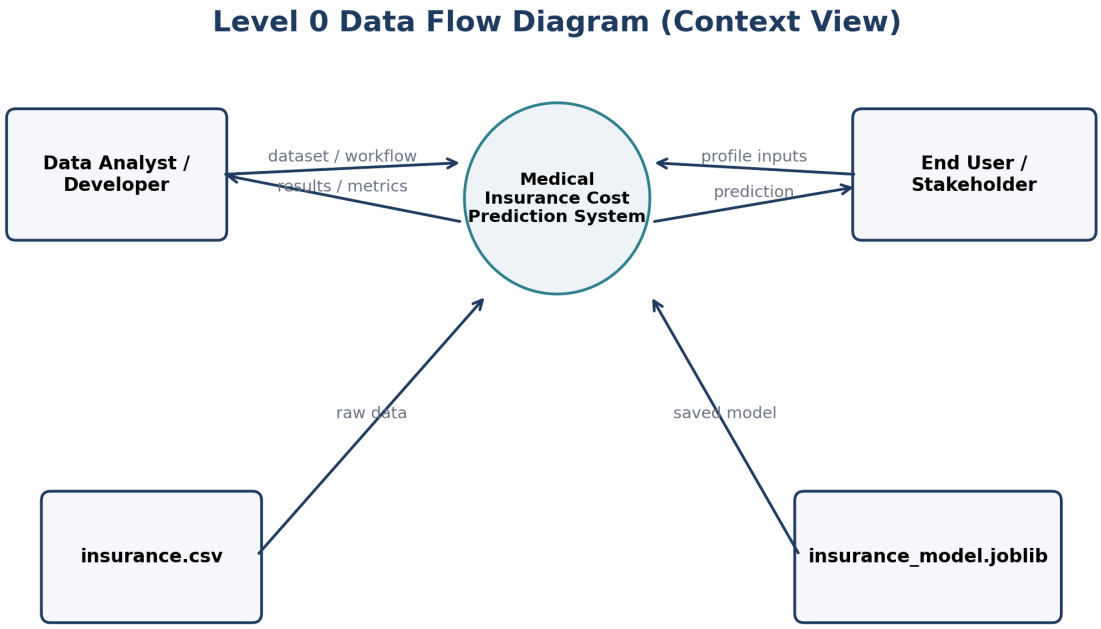


Figure 4.2: Level 0 data flow diagram

The DFD intentionally reflects the actual implementation: model training is performed in the notebook workflow, whereas app.py loads the serialized artifact and sends raw user data directly to the prediction pipeline.

4.3 ER Diagram

The project does not use a database, so the ER diagram is a logical representation rather than a physical database schema. It describes the entities that naturally arise from the data and machine learning workflow: a policyholder profile, an insurance record, a model artifact and a prediction result. This representation helps explain the information relationships without implying that a database was implemented.

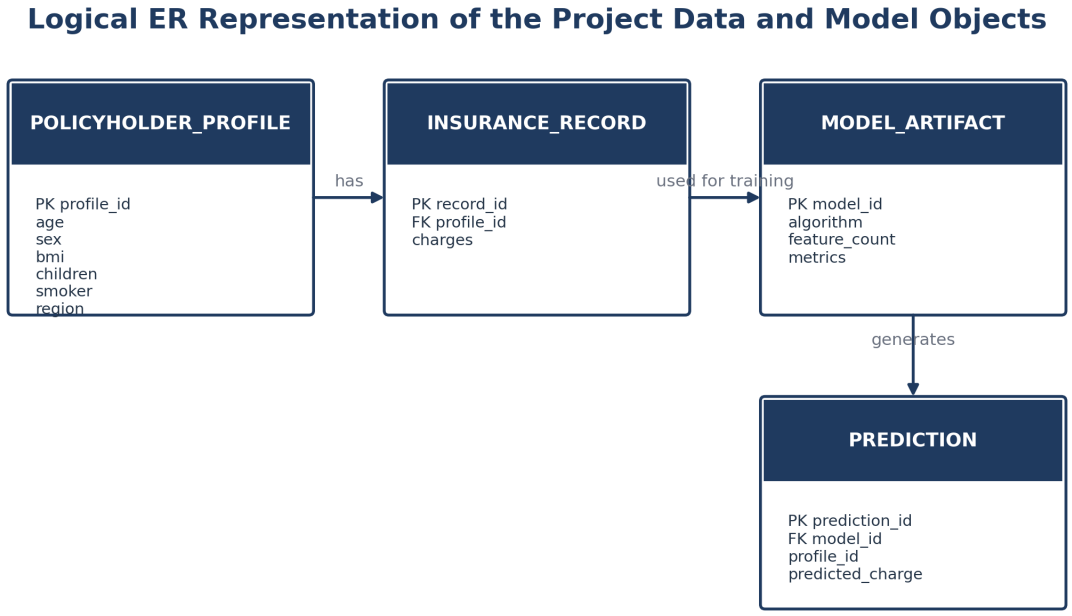


Figure 4.3: Logical ER representation of the project data and model objects

A policyholder profile is linked to its observed charge record. The model artifact stores the trained algorithm and its learned parameters. Each prediction references the model and the input profile from which the estimate was generated.

4.4 UML Diagrams

The class view highlights the software components used during prediction. StreamlitApp represents the user interface logic; InsurancePipeline encapsulates feature engineering, preprocessing and regression; ClinicalFeatureEngineer creates the interaction term; ColumnTransformer encodes the variables; and LinearRegression stores the fitted OLS parameters. The implementation uses a compact component structure rather than a multi-service architecture.

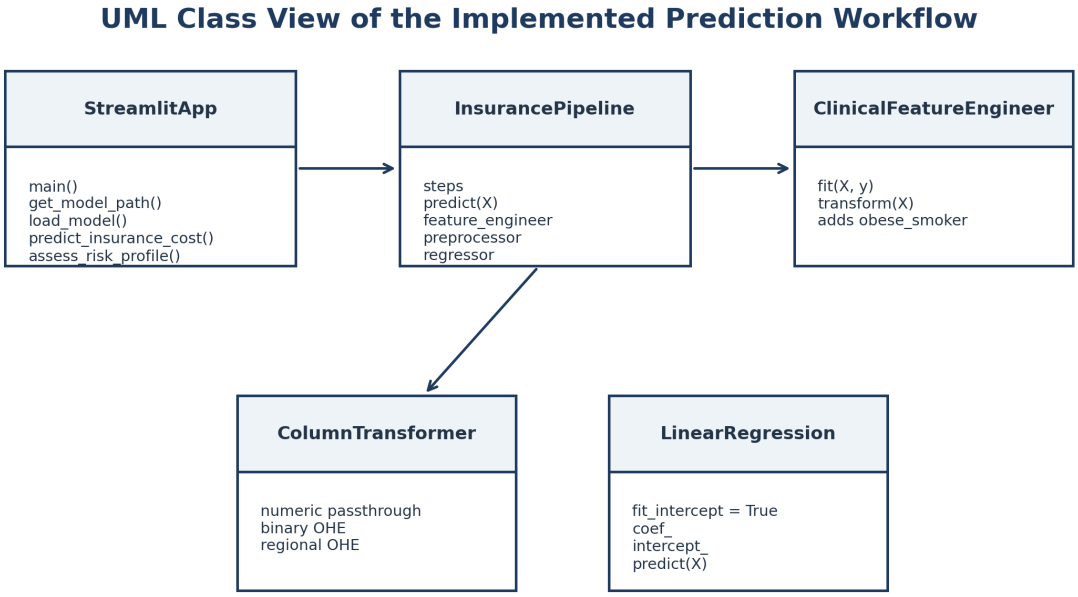


Figure 4.4: UML class view of the implemented prediction workflow

CHAPTER 5

IMPLEMENTATION

5.1 Tools and Technologies Used

Tool / Library	Purpose
Python	Primary programming language for data analysis, modelling and application logic.
Jupyter Notebook	Interactive environment for EDA, feature engineering and model evaluation.
Pandas	CSV loading, data inspection, grouping and tabular manipulation.
NumPy	Numerical operations and metric calculations.
Matplotlib	Publication-quality plots for EDA and diagnostics.
Scikit-learn	Preprocessing, OLS Linear Regression, feature transformation and evaluation metrics.
Joblib	Serialization and loading of the trained pipeline.
Streamlit	Interactive user interface for real-time inference.

Table 5.1: Tools and technologies

5.2 Algorithm: Ordinary Least Squares Linear Regression

The final estimator is Ordinary Least Squares (OLS) Linear Regression. It fits a linear function that minimizes the sum of squared residuals between observed and predicted charges. After preprocessing, the model estimates one coefficient for each of the nine final inputs: age, bmi, children, obese_smoker, sex_male, smoker_yes, region_northwest, region_southeast and region_southwest.

The fitted equation can be written as:

```
ŷ = β0 + β1(age) + β2(bmi) + β3(children) + β4(obese_smoker) + β5(sex_male) +
β6(smoker_yes) + β7(region_northwest) + β8(region_southeast) +
β9(region_southwest)
```

5.3 Feature Engineering and Preprocessing

Exploratory analysis showed that the most pronounced separation occurs when smoking and BMI are considered together. The project therefore creates a binary interaction feature instead of expanding the model into a large polynomial feature space. The rule is row-wise and deterministic.

```
df['obese_smoker'] = ((df['bmi'] >= 30.0) & (df['smoker'] == 'yes')).astype(int)
```

The interaction is implemented by ClinicalFeatureEngineer, a stateless transformer referenced by the serialized model. Because the transformer only uses values from the current row, it does not need to estimate any dataset-wide parameter. In the fitted model, numeric variables are passed through without scaling. Binary categorical variables use one-hot encoding with the redundant reference level removed, while region uses first-level dropping so that northeast serves as the reference category. The resulting transformed matrix has nine inputs.

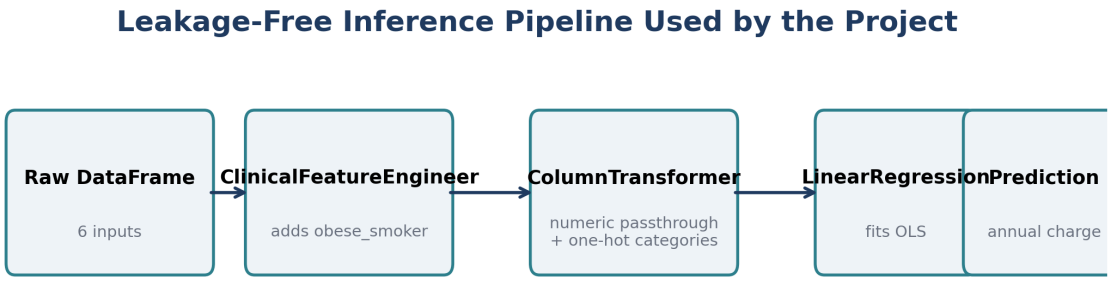


Figure 5.1: Leakage-free inference pipeline

Stage	Implementation role
Raw DataFrame	Receives the six original predictors.
ClinicalFeatureEngineer	Adds obese_smoker without fitting a numeric parameter.
ColumnTransformer	Passes numeric columns through and encodes categorical columns.
LinearRegression	Fits the OLS coefficients and intercept.
Prediction	Returns the estimated annual charge.

Table 5.2: Pipeline stages and implementation roles

5.4 Model Training and Persistence

The dataset is split into 1,070 training observations and 268 testing observations using test_size = 0.2 and random_state = 42. Two OLS models are evaluated under the same partition: a baseline six-feature model and an enhanced model that includes obese_smoker. The comparison isolates the contribution of the engineered feature without changing the underlying estimator.

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train_transformed, y_train)
predictions = model.predict(X_test_transformed)
```

The final model is stored in models/insurance_model.joblib. The serialized artifact is not just a coefficient array: it contains the feature-engineering step, the fitted ColumnTransformer and the fitted LinearRegression estimator. This allows app.py to accept the same raw six-column schema used by the modelling workflow.

Implementation Verification

The fitted artifact contains three logical steps named feature_engineer, preprocessor and regressor. Independent reproduction from insurance.csv produced the same holdout metrics as the project notebook and recovered the same coefficient values and sample prediction. This confirms that the modelling numbers reported in this document are consistent with the supplied dataset and serialized estimator.

5.5 Streamlit Application

The user interface is implemented in app.py and is designed around direct pipeline ingestion. The application collects age, biological sex, number of children, BMI, smoking status and residential region. These values are assembled into a Pandas DataFrame with the canonical raw schema and passed directly to the saved pipeline.

- Wide two-column layout separating profile inputs from the estimated expenditure output.
- Age, children and BMI sliders plus radio buttons for sex and smoking status and a region selector.
- Currency-formatted predicted annual charge presented as the primary output.
- Benchmark indicators comparing the estimate with the dataset median and mean.
- Risk-context messaging based on BMI category and smoker/obese-smoker combinations.
- An explicit disclaimer stating that the tool is for educational/informational use and is not an actual insurance quote.

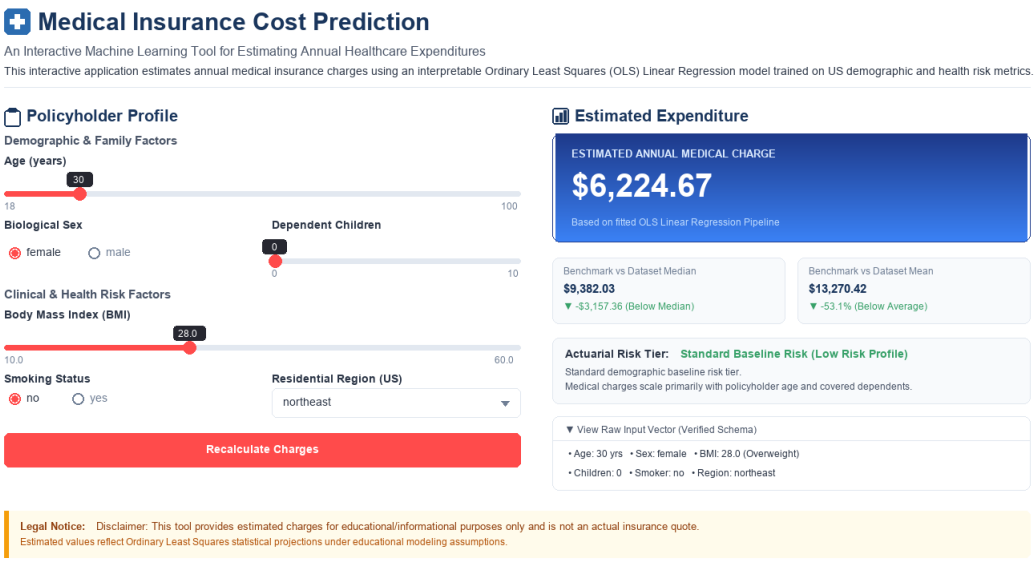


Figure 5.2: Streamlit interface layout derived from app.py

The application is intended for local interactive demonstration. It does not use a database, account system or external insurance API, and it should not be interpreted as a production underwriting service.

CHAPTER 6

RESULTS AND DISCUSSION

6.1 Exploratory Data Analysis

The target variable charges has a pronounced positive skew. The sample mean is USD 13,270.42 while the median is USD 9,382.03. The gap between these central measures arises from a relatively small high-cost tail that extends to USD 63,770.43. Such a distribution makes subgroup analysis particularly informative because a few high-charge observations can influence both the mean and regression loss.

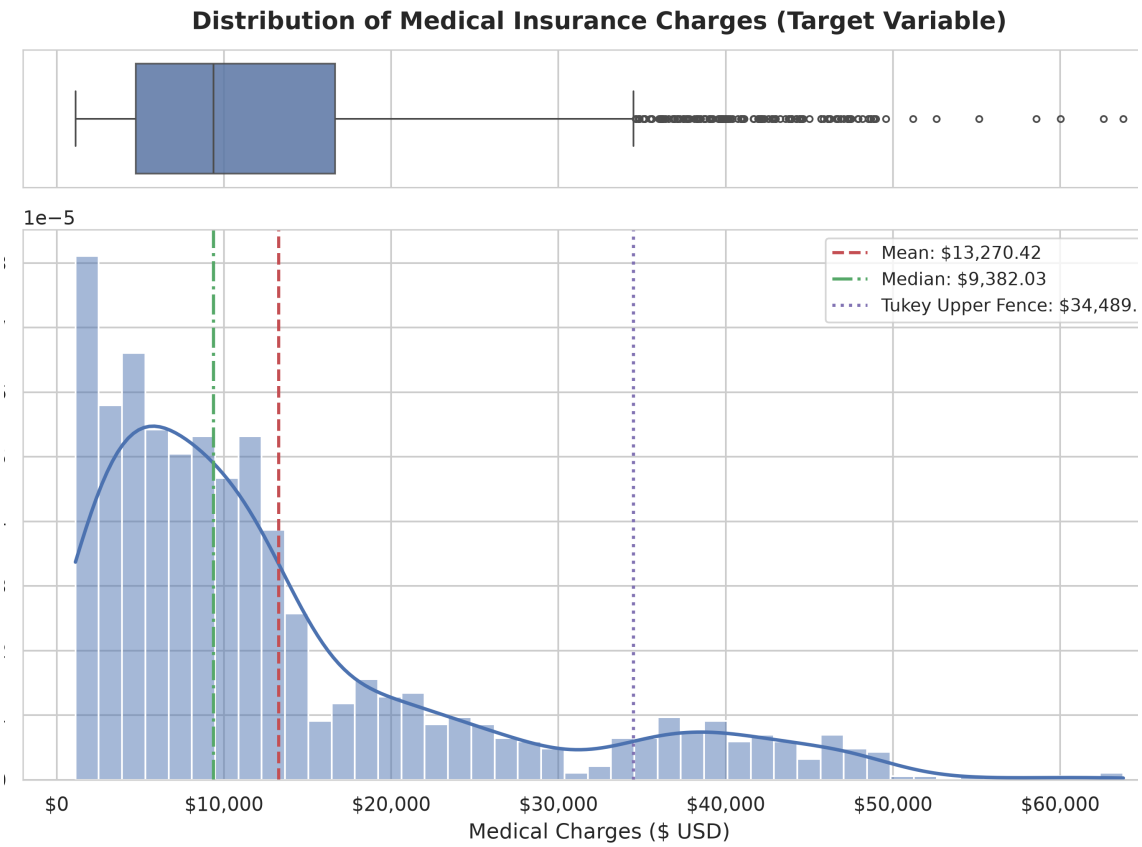


Figure 6.1: Distribution of annual medical insurance charges

Age shows a positive empirical association with charges for both smokers and non-smokers. The two groups occupy visibly different charge bands, indicating that age alone is insufficient to represent the major separation in the target.

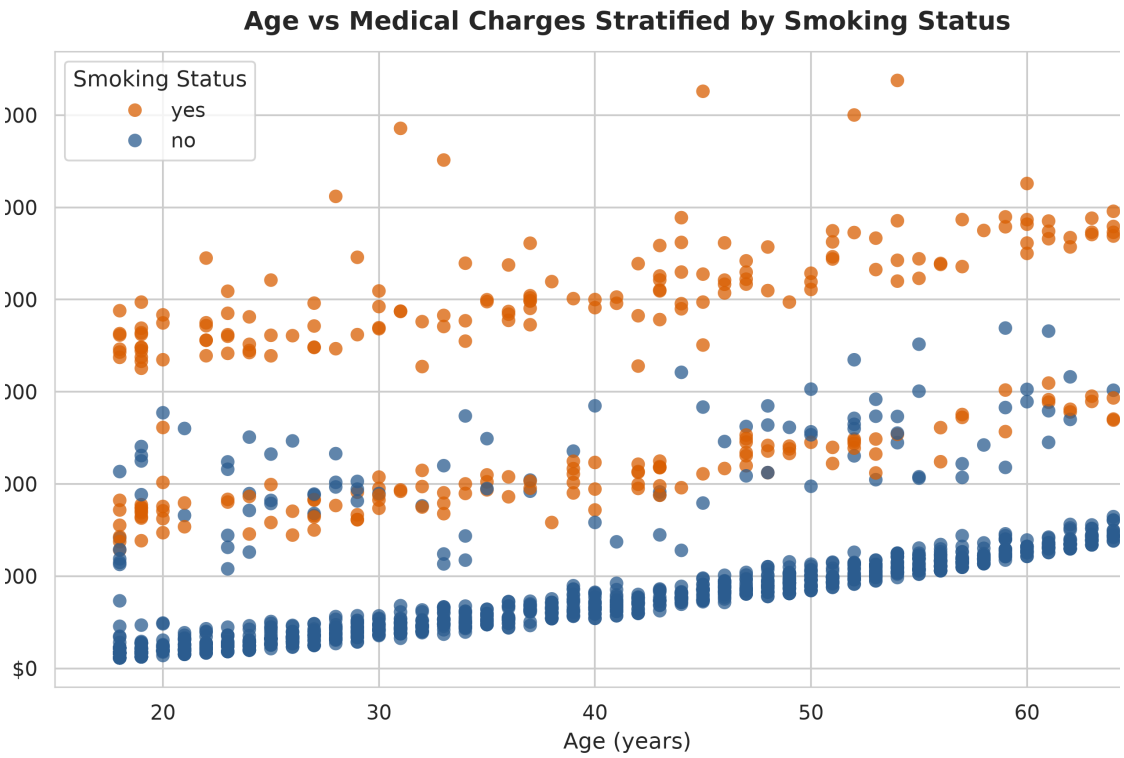


Figure 6.2: Age vs medical charges by smoking status

6.1 Exploratory Data Analysis (continued)

BMI displays a comparatively modest relationship with charges among non-smokers, but the upper part of the smoker distribution becomes substantially more expensive around and above the BMI 30 threshold used by the project to define the engineered obesity indicator. This visual discontinuity motivated the obese_smoker feature.

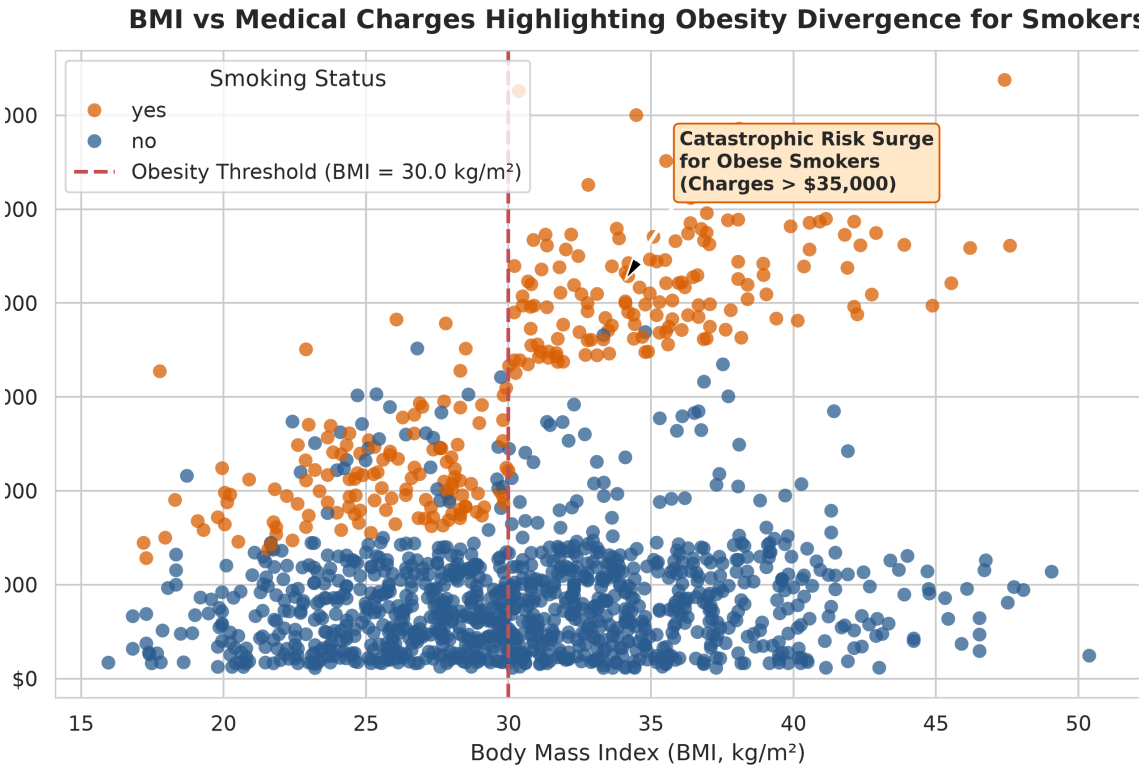


Figure 6.3: BMI vs medical charges and the smoker-obesity interaction

Smoking status is the clearest single binary separator in the dataset. The mean charge for smokers is USD 32,050.23 compared with USD 8,434.27 for non-smokers, an empirical mean difference of USD 23,615.96.

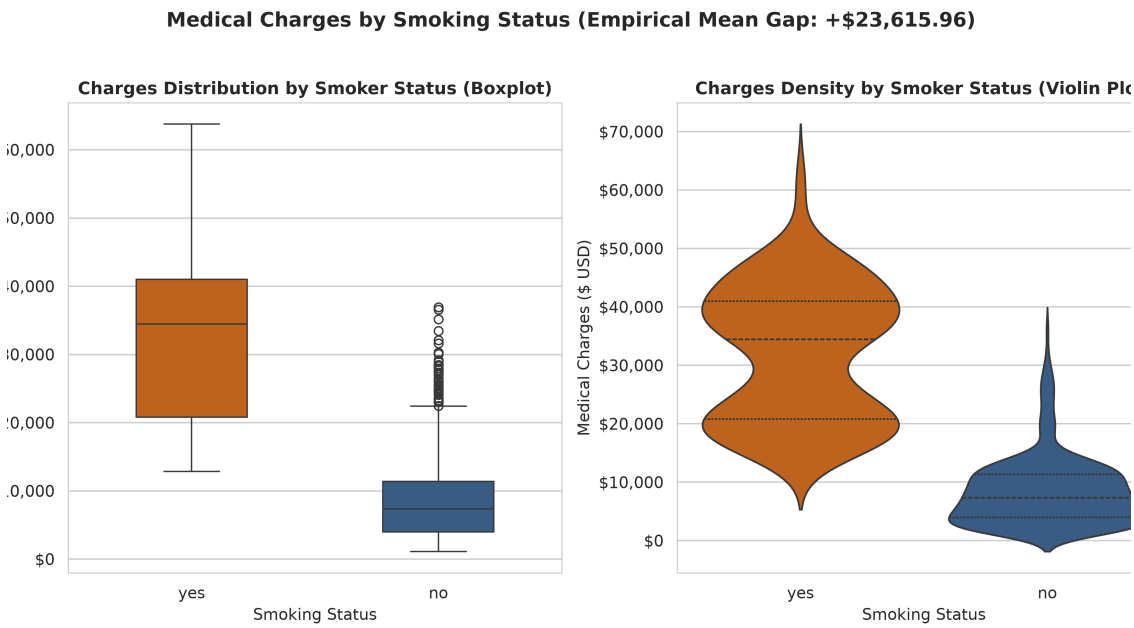


Figure 6.4: Medical charges by smoking status

Secondary Demographic and Correlation Analysis

The remaining demographic variables show smaller group differences. Sex medians are close even though the male sample mean is somewhat higher. Charges vary across dependent counts, but the groups with four or five children are small, so their sample means should be interpreted cautiously. Regional mean charges differ, with the Southeast being the highest of the four regions in the sample.

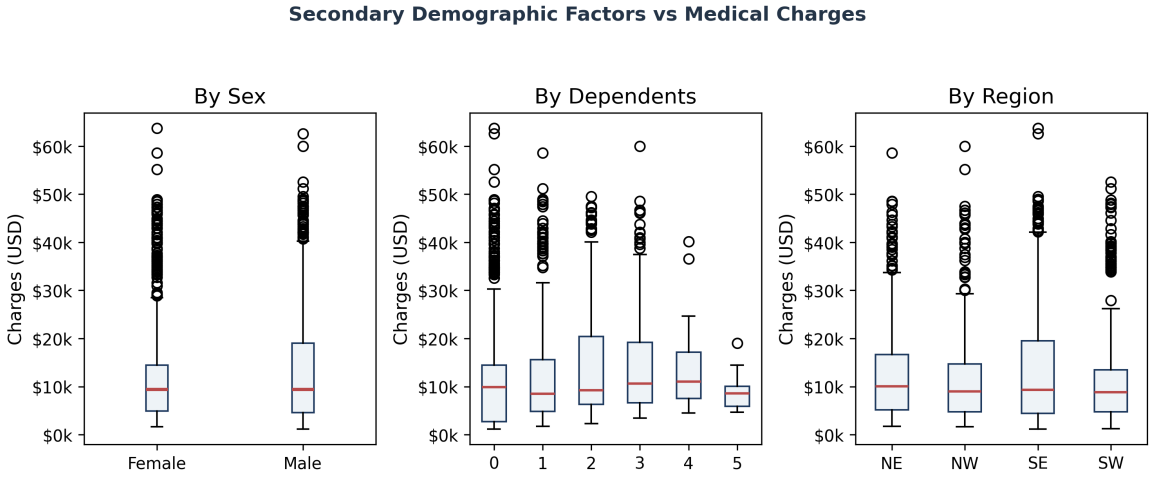


Figure 6.5: Secondary demographic factors vs medical charges

The Pearson correlation analysis shows that obese_smoker has the strongest linear correlation with charges (about 0.81), followed by smoker_yes (about 0.79), age (about 0.30) and bmi (about 0.20). The weak inter-feature correlations also indicate that severe multicollinearity is not visible among the main predictors at the raw-feature level.

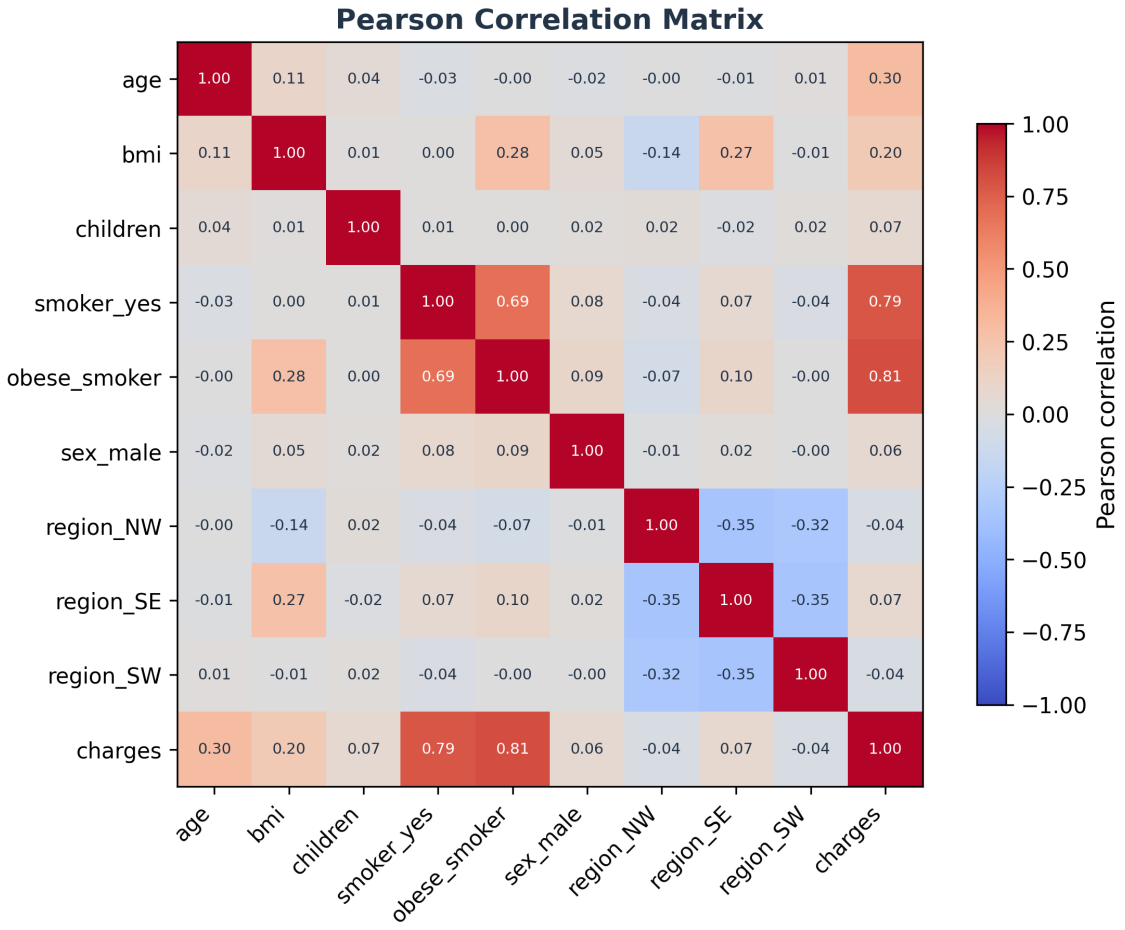


Figure 6.6: Pearson correlation matrix

6.2 Outlier Analysis

Tukey's IQR rule is used as a diagnostic rather than an automatic deletion rule. The first quartile of charges is USD 4,740.29 and the third quartile is USD 16,639.91, giving an IQR of USD 11,899.63 and an upper fence of USD 34,489.35. A total of 139 of 1,338 observations lie above that threshold.

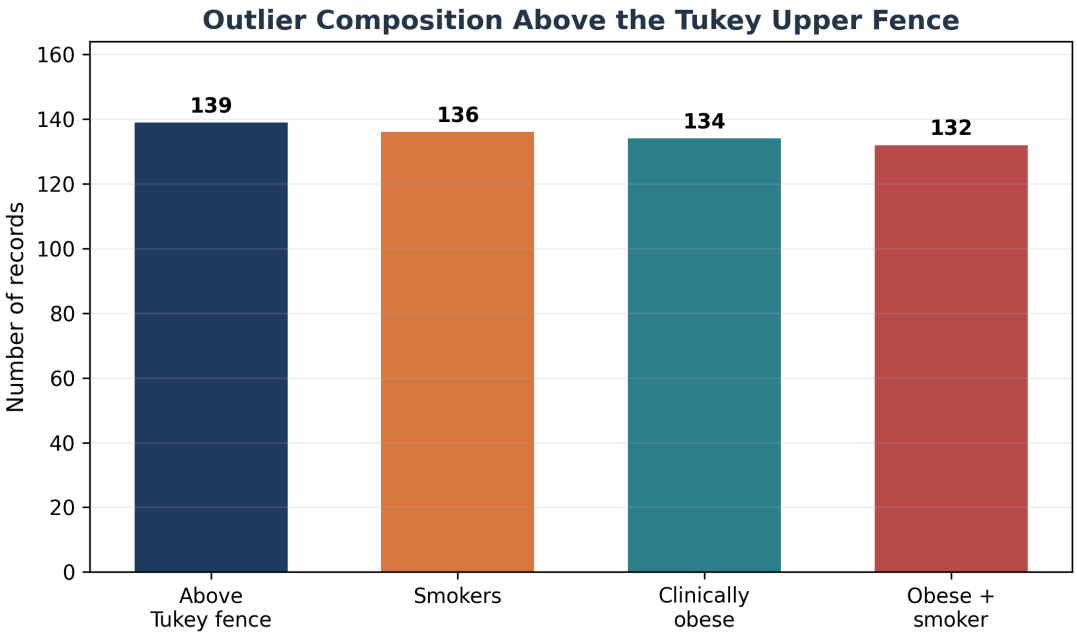


Figure 6.7: Outlier composition above the Tukey upper fence

Measure	Value
Records above upper fence	139 of 1,338 (10.39%)
Outlier records that are smokers	136 of 139 (97.84%)
Outlier records that are obese	134 of 139 (96.40%)
Outlier records that are obese smokers	132 of 139 (94.96%)
Total dataset charge sum	$17,755,824.99
Charge sum above upper fence	$5,852,448.66 (32.96%)

Table 6.1: Quantitative profile of flagged high-charge records

The project retains these observations. The dataset provides no evidence that high charges are data-entry errors, and the exact clinical cause of any particular high charge cannot be inferred because diagnoses and treatment records are not included. Removing the upper tail would discard a substantial share of the observed charge liability and could make the model less representative of high-cost cases.

6.3 Model Performance

The two models use the same train-test split, the same target and the same categorical representation. The enhanced model differs only through the addition of obese_smoker. On the 268-record holdout set, the enhanced OLS model materially reduces all error measures and increases the explained variance.

Metric	Baseline OLS	Enhanced OLS	Change
MAE	$4,181.19	$2,486.32	40.5% lower error
MSE	33,596,915.85	19,876,213.86	40.8% lower
RMSE	$5,796.28	$4,458.27	23.1% lower
R^2	0.7836	0.8720	+0.0884 points

Table 6.2: Holdout test performance of baseline and enhanced OLS models

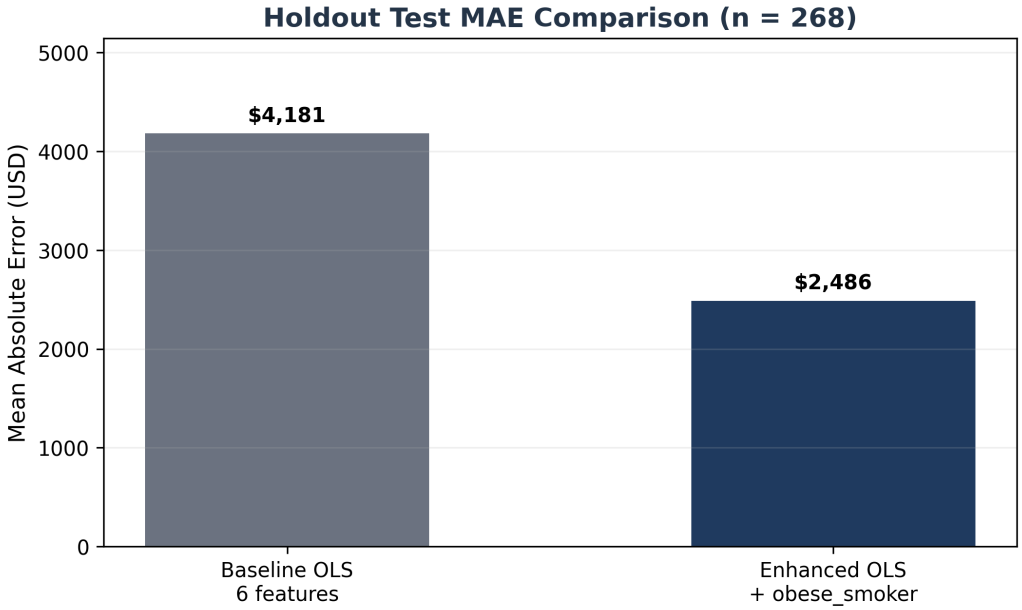


Figure 6.8: Holdout test MAE comparison

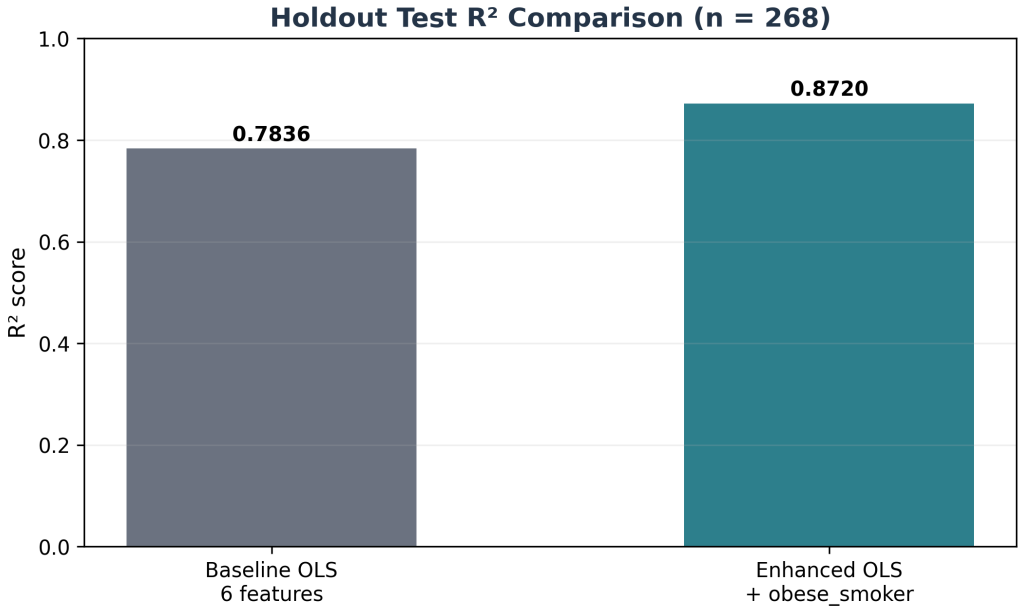


Figure 6.9: Holdout test R^2 comparison

The improvement is not caused by changing to a more complex estimator. It comes from providing the linear model with one additional variable that explicitly represents the strongest combined pattern identified during EDA.

6.4 Model Coefficients and Non-Causal Interpretation

The enhanced model remains fully interpretable because each transformed feature has a directly inspectable coefficient. The coefficient descriptions below are associative, not causal: they describe the model's fitted relationships with the other predictors held constant under the linear specification.

Feature	Coefficient (USD)	Model interpretation
age	+261.81	Associated with about USD 261.81 higher predicted charge per additional year, holding the other model inputs constant.
bmi	+56.58	Associated with about USD 56.58 higher predicted charge per one-unit BMI increase in the baseline/non-smoker component.
children	+469.79	Associated with about USD 469.79 higher predicted charge per additional covered dependent.
obese_smoker	+19,850.74	Additional model adjustment for the combined condition of BMI >= 30 and smoker = yes.
sex_male	-502.42	Model difference for male relative to the female reference category.
smoker_yes	+13,292.21	Model difference for smoker relative to the non-smoker reference when obese_smoker = 0.
region_northwest	-339.00	Regional adjustment relative to the northeast reference.
region_southeast	-692.16	Regional adjustment relative to the northeast reference.
region_southwest	-1,371.81	Regional adjustment relative to the northeast reference.

Table 6.3: Learned coefficients of the enhanced OLS model

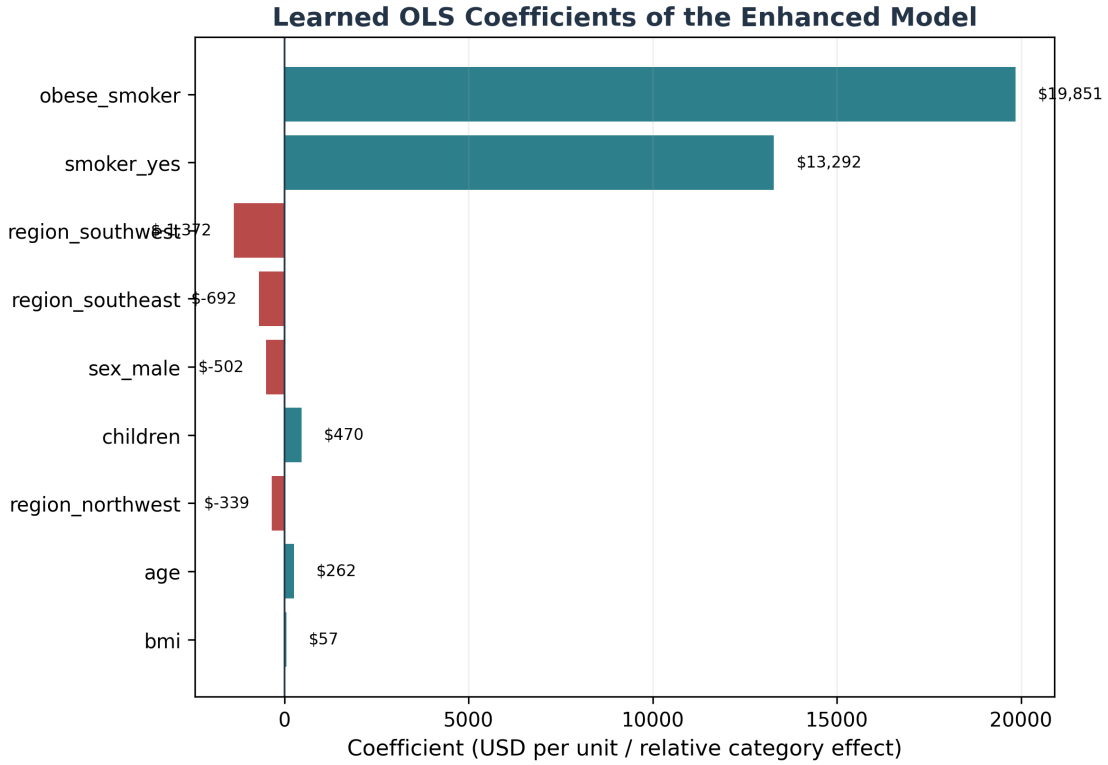


Figure 6.10: Learned OLS coefficients of the enhanced model

The fitted intercept is approximately -USD 3,213.86. The intercept should not be interpreted as a realistic standalone customer charge because the reference combination of all categorical variables and the zero point of the continuous inputs does not correspond to a typical policyholder.

6.5 Model Diagnostics

The actual-versus-predicted plot shows that predictions track the 45-degree identity line reasonably well for a broad portion of the holdout set. The Pearson correlation between the true and predicted holdout charges is approximately 0.934. The model nevertheless exhibits larger deviations among some of the most expensive cases, which is expected from a simple linear representation of a heavy-tailed cost target.

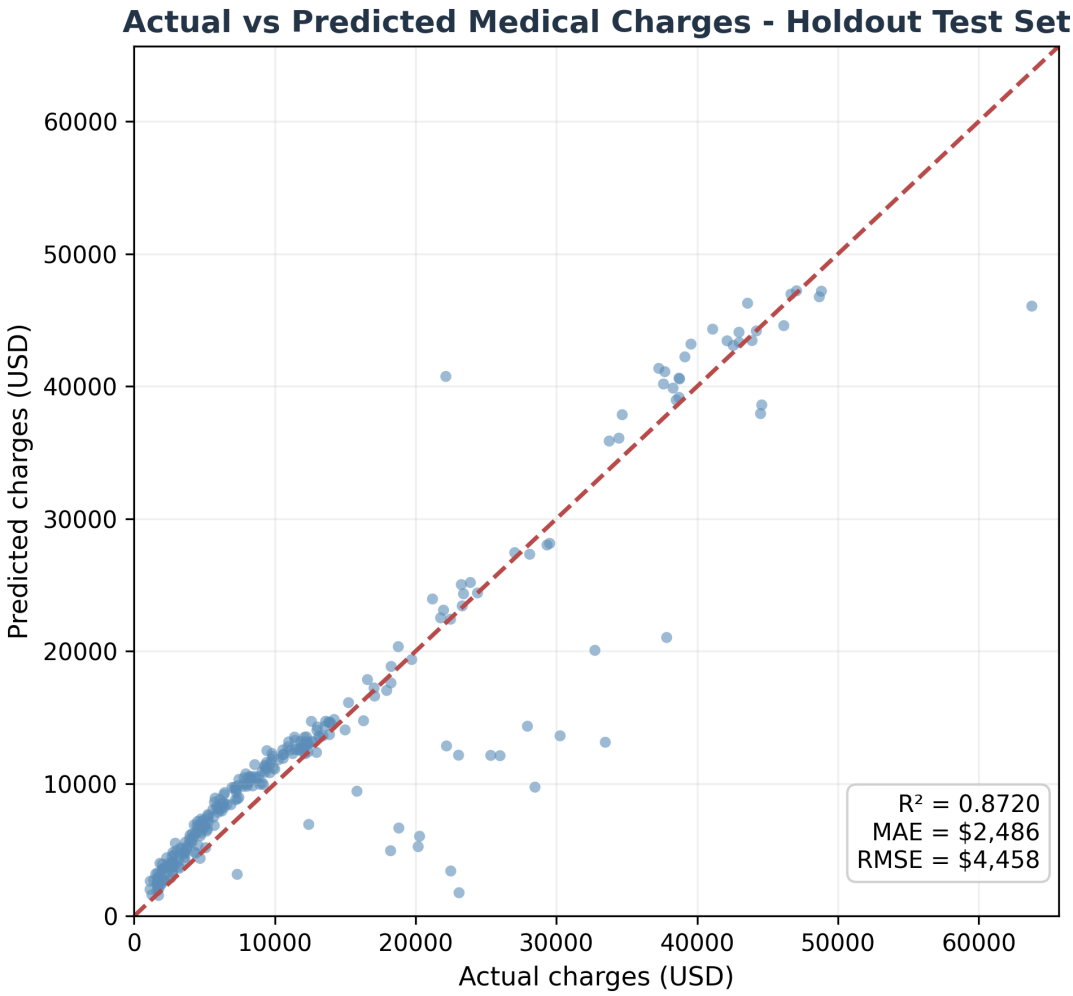


Figure 6.11: Actual vs predicted medical charges - holdout test set

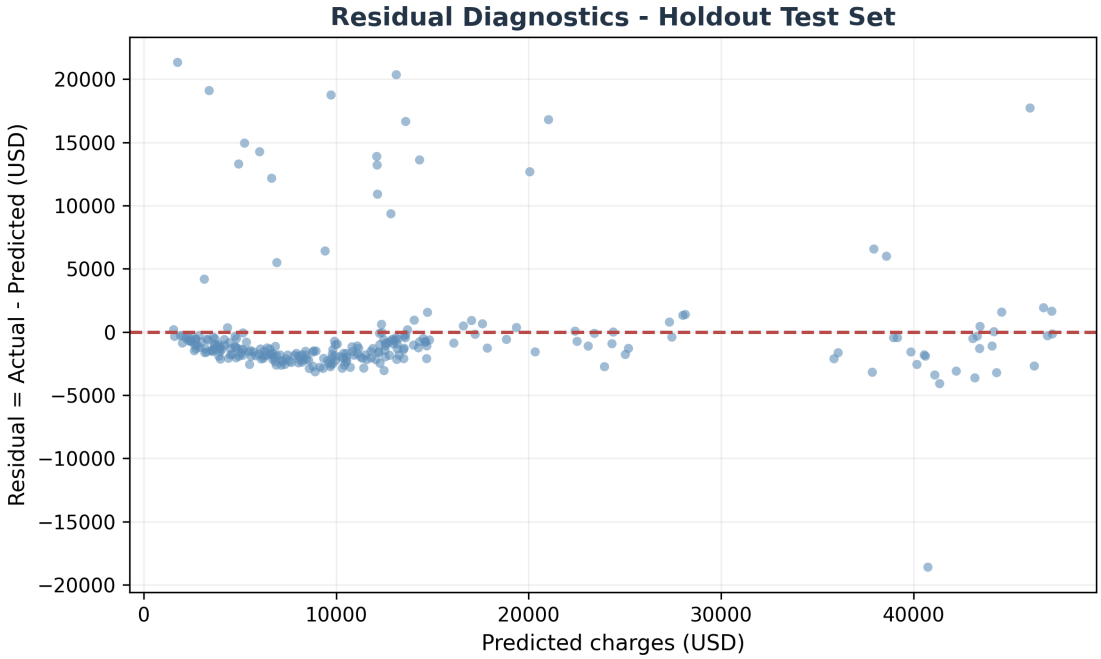


Figure 6.12: Residual diagnostics on the holdout test set

The residual mean is approximately -USD 226.55 and the sample residual standard deviation is about USD 4,460.85. The residual pattern is broadly centered on zero, although dispersion increases for higher predicted charges. This indicates that the model captures the main structure but does not eliminate the intrinsic variability of high-cost observations.

6.6 Prediction Demonstration on New Profiles

The saved model can be applied directly to a raw profile. A demonstration profile used throughout the project is a 45-year-old male with BMI 32.5, two children, smoker = yes and region = southeast. The serialized model produces an estimated annual charge of approximately USD 43,294.37 for this profile.

Scenario	Age	BMI	Smoker	Predicted charge
Baseline	30	28.0	No	$6,225
Smoker only	30	28.0	Yes	$19,517
Obese non-smoker	30	32.0	No	$6,451
Obese smoker	30	32.0	Yes	$39,594

Table 6.4: Prediction scenarios for a common baseline profile

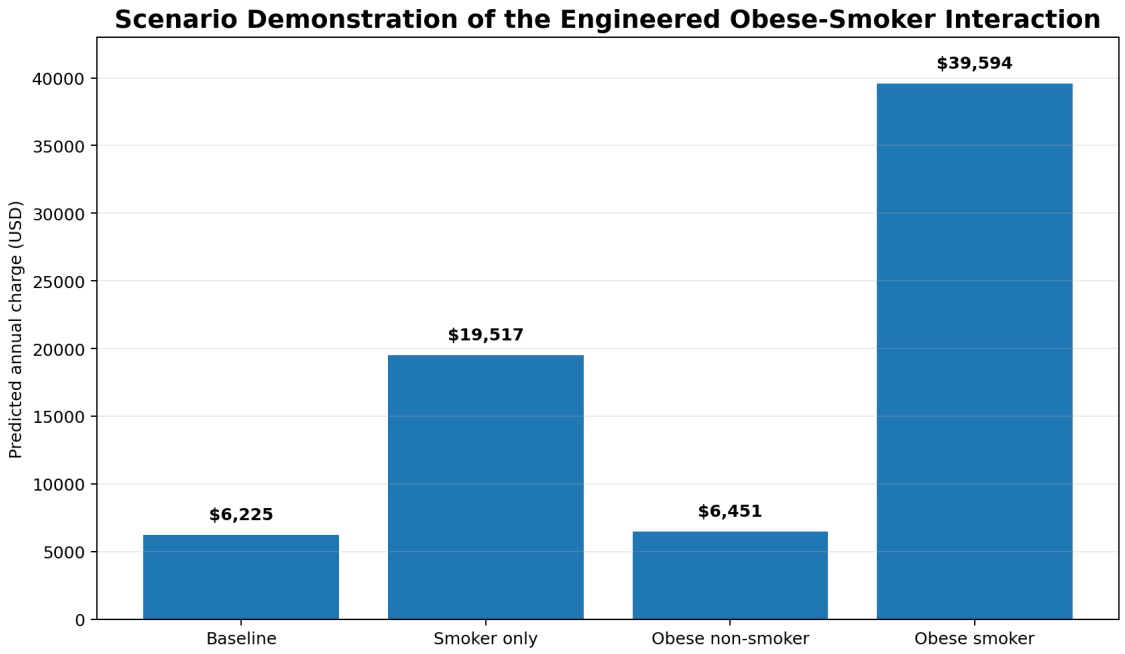


Figure 6.13: Scenario demonstration of the engineered obese-smoker interaction

The scenario analysis illustrates the purpose of the engineered feature. The obese-smoker profile receives a much larger estimate than either the smoker-only or obese-only profile, because the fitted model contains an explicit additional obese_smoker coefficient.

6.7 Discussion and Limitations

Discussion

The strongest result of the project is methodological rather than simply numerical: understanding the data made it possible to improve a transparent estimator without replacing it with a black-box model. The increase from $R^2 = 0.7836$ to $R^2 = 0.8720$ demonstrates that a targeted interaction can account for a substantial share of the previously unexplained variation in this dataset.

Limitations

- The dataset contains only 1,338 observations and six raw predictors, so the model cannot represent information that is not present in the source data.
- The target is a billed medical charge, not a clinical outcome, and the dataset does not identify diagnoses, procedures, admission history or medication use.
- The final estimator is linear, so it can miss other nonlinear relationships not explicitly represented by the engineered feature.
- The application exposes input ranges wider than the training data for some controls; users should therefore avoid treating extreme extrapolated values as reliable real-world estimates.
- The interface is an educational tool. The disclaimer in app.py makes clear that the result is not an actual insurance quote.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Summary of the Project

This project developed a complete machine learning workflow for predicting annual medical insurance charges from six demographic and lifestyle attributes. The analysis began with data validation and exploratory statistics, identified a strong smoker-related cost separation and then focused on the additional divergence among obese smokers. The engineered obese_smoker feature was incorporated into an OLS pipeline and evaluated against a baseline model under the same holdout partition.

7.2 Key Achievements

Achievement	Outcome
Data audit	1,338 records; zero missing values; one duplicate pair identified and retained.
Exploratory analysis	Right-skewed target and strong smoker/obesity interaction documented.
Feature engineering	obese_smoker added as a deterministic row-wise interaction.
Predictive improvement	R^2 increased from 0.7836 to 0.8720; MAE fell from USD 4,181.19 to USD 2,486.32.
Interpretability	Nine final model inputs and explicit OLS coefficients are directly inspectable.
Application	Streamlit interface created for real-time raw-input inference with an educational disclaimer.

Table 7.1: Key project achievements

Conclusion

The project demonstrates that a carefully specified and interpretable linear model can achieve strong predictive performance on this tabular medical-cost dataset. The result depends less on algorithmic complexity than on accurately identifying the structure of the data, representing the important interaction and evaluating the model honestly on unseen observations.

7.3 Future Enhancements

1. Expand the dataset with longitudinal policy records, richer clinical variables, procedure history and utilization measures so that the model can represent more of the true variation in medical expenditure.
2. Evaluate alternative interpretable approaches such as generalized linear models with appropriate distributions and link functions for positive, right-skewed cost targets.
3. Add uncertainty or prediction intervals so that a user receives a range rather than a single point estimate.
4. Introduce automated data validation and schema checks before training and inference to detect malformed input files or unsupported categories.
5. Package the model behind a small API and deploy the Streamlit interface to a managed environment after adding appropriate authentication, logging and monitoring.
6. Periodically re-evaluate the model for drift when new data become available, rather than assuming that historical relationships remain stable indefinitely.
7. Conduct fairness and governance reviews before any real-world decision-support use, especially because demographic and lifestyle predictors can have significant social consequences.

Closing Statement

Future development should preserve the project's main strength - a clear connection between data, modelling assumptions, measurable performance and explainable outputs. Greater model complexity should be introduced only when it produces a meaningful benefit that justifies the additional governance and interpretability cost.

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ANNEXURE A: IET LEARNING OUTCOMES

1. Apply knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the conceptualization of engineering models.

Response: The project applies regression mathematics, feature representation and error metrics to formulate and evaluate a medical-cost prediction model.

2. Identify, formulate, research literature and solve complex engineering problems using substantiated conclusions.

Response: The problem is formulated as supervised regression, supported by literature on statistical learning and healthcare cost prediction.

3. Design solutions for complex engineering problems that meet specified needs with appropriate safety and societal considerations.

Response: The solution combines an interpretable OLS pipeline with a user-facing application and an explicit educational disclaimer.

4. Conduct investigations including analysis and interpretation of data and synthesis of valid conclusions.

Response: EDA, correlation analysis, outlier diagnostics and holdout evaluation are used to support the modelling decisions.

5. Create, select and apply appropriate techniques, resources and modern engineering and IT tools.

Response: Python, Pandas, NumPy, Matplotlib, scikit-learn, joblib and Streamlit are used across the project workflow.

6. Function effectively as an individual and as a member or leader in diverse teams.

Response: The project was developed as an individual academic project with guidance from the project mentor and support from the academic environment.

7. Communicate effectively on complex engineering activities with the engineering community and society.

Response: The project is documented in a structured report and demonstrated through an interactive Streamlit interface.

8. Demonstrate understanding of societal, health, safety, legal and cultural issues and responsibilities.

Response: The report avoids causal health claims and clearly states that the application is not an actual insurance quote.

9. Understand and commit to professional ethics and responsibilities.

Response: The project emphasizes reproducibility, transparency, honest evaluation and careful interpretation of model outputs.

10. Understand the impact of engineering solutions in a societal context and the need for sustainable development.

Response: The report recognizes that predictive systems in healthcare can influence decisions and therefore require responsible interpretation and governance.

11. Demonstrate knowledge and understanding of management and business practices.

Response: The workflow addresses data quality, model risk, reproducibility, portability and future monitoring considerations.

12. Recognize the need for and engage in independent and life-long learning.

Response: The project integrates statistical learning, software implementation, visualization and deployment concepts into a single practical workflow.